Novel Views of a Person from 125 Photos: NeRF, TensoRF, 3DGS and Mip-Splatting

Anas Khan MBZUAI CV802 Sub-project 3: Novel-view Synthesis



Figure 1. Three of the 16 held-out photos (not used in training) next to each method's render from the same camera.

Method	PSNR↑	SSIM↑	LPIPS↓	Train (min)	FPS↑	Primitives	MB
NeRF [1]	26.59	0.931	0.132	60	0.04	1.2M	14
TensoRF [2]	26.35	0.938	0.130	42	0.98	17.4M	67
3DGS [3]	26.63	0.951	0.072	**16**	373	149k	35
Mip-Splatting [4]	**27.62**	**0.958**	**0.067**	31	**569**	197k	47

Table 1. Held-out photos (16, 750×1000): PSNR [dB], SSIM [5], LPIPS (VGG) [6], training minutes (runs shared one RTX 5000 Ada), images per second, primitives (Gaussians or network/grid values), MB. Best in bold.

Rendered at	3DGS	Mip-Splatting
1/4 size (zoom out)	23.71 / 0.055	**28.63** / 0.028
1× (training size)	26.63 / 0.072	**27.62** / 0.067
1.6× (zoom in)	26.29 / 0.090	**27.37** / 0.084

Table 2. Zoom test: PSNR / LPIPS rendered at other sizes. Truth: photo averaged down ($\frac{1}{4}$) or the 1600 px original (1.6×).

1 Introduction

The task is to render a person from *new* viewpoints (preset-path video, interactive viewer). Input: only the 125 photos (four iPhone lenses, video frames) and their cameras, recovered in Sub-project 1 with VGGSfM [7] and undistorted; no point cloud is used, and the Gaussian methods start from 100k random points. Person masks give a white background; 1000 px. Every 8th photo (16) is held out, as in [3].

2 Methods

NeRF [1] and TensoRF [2] render a pixel by summing density σ_i and colour c_i along its ray, $C = \sum_i T_i(1-e^{-\sigma_i\delta_i})c_i$, $T_i = \prod_{j<i} e^{-\sigma_j\delta_j}$. NeRF asks one MLP (8×256, positional encoding, coarse then fine samples); TensoRF a 3D feature grid factorised into planes times lines (VM decomposition) and a tiny MLP. **3D Gaussian Splatting [3]** stores the scene as 3D Gaussians (position, covariance Σ, opacity, spherical-harmonics colour), grown from the random points while training. Each is projected to a 2D Gaussian $\Sigma' = JW\Sigma W^\top J^\top$ (EWA splatting [8]), sorted by depth and alpha-blended: real time. **Mip-Splatting [4]** adds a 3D smoothing filter (no Gaussian finer than the photos sample) and a 2D Mip filter instead of 3DGS's fixed screen-space dilation, so renders at other zoom levels do not alias.

All use the authors' code (NeRF: nerf-pytorch, the PyTorch port of the original), white background, 30k iterations; NeRF 1 h (44k steps; the paper 200k+). Extra: 3DGS with the room (17.96 dB).

3 Results

Table 1, Fig. 1: Mip-Splatting is the most faithful (27.62 dB, LPIPS 0.067). 3DGS (26.63) ties NeRF (26.59) and TensoRF (26.35 dB) in PSNR but is far sharper (LPIPS 0.072 vs 0.132, 0.130), and the Gaussian methods draw at 373–569 images per second while the ray marchers need seconds per image. Table 2 tests the claim of [4]: at a quarter of the size 3DGS falls to 23.71 dB, its Gaussians thinner than a pixel, while Mip-Splatting keeps 28.63 dB. Ablation: starting from the Sub-project 1 point cloud instead of random points gives 27.53 (3DGS), 27.53 (Mip-Splatting) and 19.71 dB (room): it helps 3DGS and the room, not Mip-Splatting.

4 Conclusion

(1) Gaussian methods suit this task: they preserve fine detail such as the shirt print and the face and render in real time, whereas the ray-marching methods produce softer images at a second or more per image. **(2) Mip-Splatting is the best choice:** it leads on every metric and at every zoom level. **(3) Images and cameras suffice:** from random initialisation every method learns the person; a point cloud helps only where views are scarce, as for the room. **(4) Coverage and training time are the limits:** regions that no photo observes cannot be recovered, and NeRF requires far more than one hour of training.

References

[1] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng, "NeRF: Representing scenes as neural radiance fields for view synthesis," in *European Conference on Computer Vision (ECCV)*, 2020.

[2] A. Chen, Z. Xu, A. Geiger, J. Yu, and H. Su, "TensoRF: Tensorial radiance fields," in *European Conference on Computer Vision (ECCV)*, 2022.

[3] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, "3D Gaussian Splatting for real-time radiance field rendering," *ACM Transactions on Graphics*, vol. 42, no. 4, 2023.

[4] Z. Yu, A. Chen, B. Huang, T. Sattler, and A. Geiger, "Mip-Splatting: Alias-free 3D Gaussian Splatting," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[5] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, 2004.

[6] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.

[7] J. Wang, N. Karaev, C. Rupprecht, and D. Novotny, "VGGSfM: Visual geometry grounded deep structure from motion," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[8] M. Zwicker, H. Pfister, J. van Baar, and M. Gross, "EWA volume splatting," in *IEEE Visualization*, 2001.